

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 17)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

01 . 2

Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

01 . 3

Explain **three** ways in which coral reefs are important to humans.

[3 marks]

1

.....

2

.....

3

.....

10

0 2 . 1

Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

EDGE species

.....

.....

.....

Endemic species

.....

.....

.....

Keystone species

.....

.....

.....

Flagship species

.....

.....

.....

0 2 . 2

Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

.....

.....

.....

.....

10

03 . 1

Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

.....

.....

.....

.....

.....

.....

03 . 2

Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

03 . 3

Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

.....

.....

.....

03 . 4

State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

[1 mark]

.....

10

0 4

Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 4

. 1

Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

0 4

. 2

Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

04 . 3

Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

.....

.....

.....

.....

04 . 4

Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

.....

.....

.....

.....

10

05

Table 1 lists some feedback mechanisms that affect the Earth's climate.

05 . 1

Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	Type of feedback	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO ₂ levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

.....

.....

.....

.....

.....

.....

0 5 . 2

Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

0 5 . 3

Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

0 5 . 4

Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

10

06 . 1

Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

.....

.....

.....

.....

.....

.....

06 . 2

Describe **five** features of a suitable release site.

[5 marks]

1

.....

2

.....

3

.....

4

.....

5

.....

06 . 3

Explain the importance of post-release monitoring.

[2 marks]

.....

.....

.....

.....

10

0	7	.	1
---	---	---	---

Describe the main principles of the circular economy.

[6 marks]

0	7	.	2
---	---	---	---

Explain what is meant by 'optimum production rather than maximum production', using a comparison between a natural ecosystem and an agroecosystem.

[2 marks]

0	7	.	3
---	---	---	---

Explain why international cooperation (eg through the EU or UN) is needed to implement circular economy initiatives.

[2 marks]

10

08

A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

08

1

Use the formula $D = N(N - 1) / \sum n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

08

2

Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

5

0 9

Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 9**1**

Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 9**2**

Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 9**3**

Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

09 . 4

Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

10

10 . 1

Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

Do not write outside the box

Extra space

10 . 2

State what is meant by a **taxon**.

[1 mark]

.....

.....

10

Do not write outside the box

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
---	---	---	---

Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

1 1 . 2

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS